

APPENDIX B: PER-SPECIES METRICS

Complete Phase 3b Per-Species Classification Report

Species	Precision	Recall	F1-Score	Support
Abutilon Indicum	53.3%	61.5%	57.1%	26
Aloe barbadensis miller	90.5%	86.4%	88.4%	22
Calotropis gigantea	61.1%	61.1%	61.1%	18
Canna indica	66.7%	73.3%	69.8%	30
Cissus quadrangularis	96.0%	77.4%	85.7%	31
Curcuma longa	41.2%	87.5%	56.0%	24
Eclipta prostrate	88.2%	66.7%	75.9%	45
Eichhornia Crassipes	85.7%	100.0%	92.3%	48
Hibiscus Rosasinensis	94.4%	81.0%	87.2%	21
Ixora coccinea	78.6%	55.0%	64.7%	20
Justica adhatoda	88.9%	84.2%	86.5%	19
Murraya koenigii	64.3%	47.4%	54.5%	19
Ocimum tenuiflorum	47.8%	47.8%	47.8%	23
Ouretlanata	100.0%	80.8%	89.4%	26
Phyllanthus amarus	85.7%	60.0%	70.6%	20
Ricinus communis	66.7%	18.2%	28.6%	22
Senna Atriculata	60.6%	69.0%	64.5%	29
Sesbania grandiflora	47.1%	84.2%	60.4%	19
Trifolium Repens	0.0%	0.0%	0.0%	1
Ziziphus mauritiana	0.0%	0.0%	0.0%	0
Weighted Average	74.6%	70.8%	70.8%	463
Macro Average	65.8%	62.1%	62.0%	463

High-Performing Species (F1 > 80%): The leaders with F1 scores above 80% were six species, in the top place *Eichhornia Crassipes* (F1 92.3%, 100% recall) secured this position mainly due to its striking floating rosette morphology providing strong visual discriminability. *Cissus quadrangularis* (precision 96.0%) and *Ouretanata* (100% precision) were found to be extremely reliable when making predictions.

Mid-Performing Species (F1 50-80%): Varying precision-recall tradeoffs of nine species along with a distinct failure mode were identified. The *Curcuma longa* plant had a high recall of 87.5% but low precision of only 41.2% indicating that it was often misclassified as another similar elongated-leaf species. Meanwhile, the *Ricinus communis* plant had a moderate precision of 66.7% with a very low recall of only 18.2%, meaning the model misses most instances, suggesting both the inter-species visual similarity and the insufficient training data caused these problems.

Poor-Performing Species (F1 < 50%): The *Ocimum tenuiflorum* plant has only achieved 47.8% F1 even with 23 validation samples; suggesting that they do not have unique or clear features. The *Trifolium Repens* plant (only 1 sample) and the *Ziziphus mauritiana* plant (0 samples) are impossible classification tasks and hence were excluded in the EfficientNetB3 configuration.